

The empirical recurrence for OEIS A253310, recovered from its tail

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Abstract

OEIS A253310 records “Empirical recurrence of order 62” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 785$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 12$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{62} = 42674688 \neq 0$, so no further tail satisfies anything shorter; any order-62 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253310 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with every 2×2 subblock diagonal maximum minus antidiagonal maximum unequal to its neighbors horizontally, vertically, diagonally and antidiagonally.”. It has offset 1 and begins

23524, 43220, 266074, 516928, 2643738, 6728458, 30117394, 99695754, ...

The entry states:

Empirical recurrence of order 62 (see link above).

As of the “Last modified” line on the live entry (Sep 01 2025 04:30:26, revision 11) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 785$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 785$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 62: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 12$ is the first whose minimal order is exactly the stated 62.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1570$ exact terms from $n = 12$ on, it returns order 62 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	0	c_{17}	2053692	c_{33}	−15182398848	c_{49}	47769444032
c_2	36	c_{18}	3349413	c_{34}	−838635012	c_{50}	−558722400
c_3	2	c_{19}	582354	c_{35}	25299533976	c_{51}	−79413506816
c_4	−417	c_{20}	−2036990	c_{36}	2640206540	c_{52}	14059440576
c_5	−66	c_{21}	−9939304	c_{37}	−50464111328	c_{53}	23550069120
c_6	1606	c_{22}	27472284	c_{38}	3521391704	c_{54}	−15670215744
c_7	674	c_{23}	54110668	c_{39}	83867760960	c_{55}	−9757253760
c_8	1998	c_{24}	−106437798	c_{40}	−5981638968	c_{56}	827578752
c_9	176	c_{25}	−346813628	c_{41}	−81016564864	c_{57}	6045030144
c_{10}	−37581	c_{26}	125729253	c_{42}	−1346131296	c_{58}	−34231680
c_{11}	−53196	c_{27}	1017219700	c_{43}	60618874992	c_{59}	1364967936
c_{12}	167338	c_{28}	−109531301	c_{44}	6269225264	c_{60}	233438976
c_{13}	367406	c_{29}	−2500591808	c_{45}	−55851393376	c_{61}	0
c_{14}	−243725	c_{30}	702491108	c_{46}	−29091022240	c_{62}	42674688
c_{15}	−1145470	c_{31}	7345169200	c_{47}	42969958656		
c_{16}	−1117774	c_{32}	−1428299386	c_{48}	26857406656		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A253310 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 12$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{62} - c_1 x^{61} - \dots - c_{62}$. Its constant term is $-c_{62} = -42674688$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with

$\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 62 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 62, so $\deg q = 62$, and it is monic. It holds from some index t on. From $\max(t, 12)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 62, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 62, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 42674688.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-62 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A253310.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.